

HAT-P-37b

R-2320

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-37_210525 · created 2026-08-18T00:55:35+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459359.9438 +/- 0.005 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.171 +/- 0.015
Transit depth (Rp/Rs)^2	2.92 +/- 0.53 [%]
Orbital Inclination (inc)	89.56 +/- 1.21
Airmass coefficient 1 (a1)	1.328 +/- 0.031
Airmass coefficient 2 (a2)	-0.27 +/- 0.37
Scatter in the residuals of the lightcurve fit is	2.32 %
Best Comparison Star	#2 - [251, 96]
Optimal Method	PSF photometry
Transit Duration (day)	0.0848 +/- 0.0051

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.171 ± 0.015 vs 0.1394 (deviation 1.96σ)
Duration fit vs published	0.0848 d vs 0.0983 d (deviation 2.47σ)
Mid-time O-C	-0.5 min
Depth SNR	10.6
Residual scatter	2.32 %

Light curve

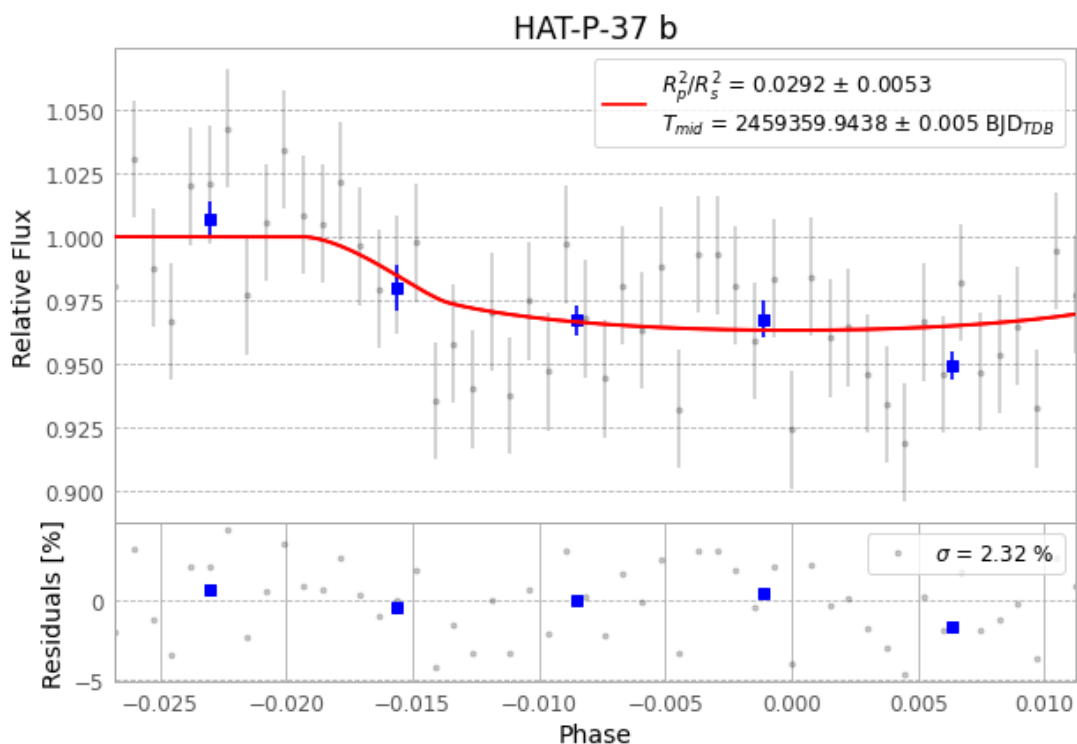


Figure 1 — FinalLightCurve_HAT-P-37 b_2021-05-25.png (SHA-256 c3f9bade2cd7bc66...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [403, 237], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b561dbc0957c7390...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

55 FITS frames (36 MB), HATP-37210525083912.FITS → HATP-37210525112112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-37-210525.log (f0b3100eab69...), FinalLightCurve_HAT-P-37 b_2021-05-25.png (c3f9bade2cd7...), FinalLightCurve_HAT-P-37 b_2021-05-25.csv (0b46355bbb71...)

Compute resources

Wall time 168.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-18 00:55Z.